

AWS S3 and Salesforce Data Cloud Integration Architecture

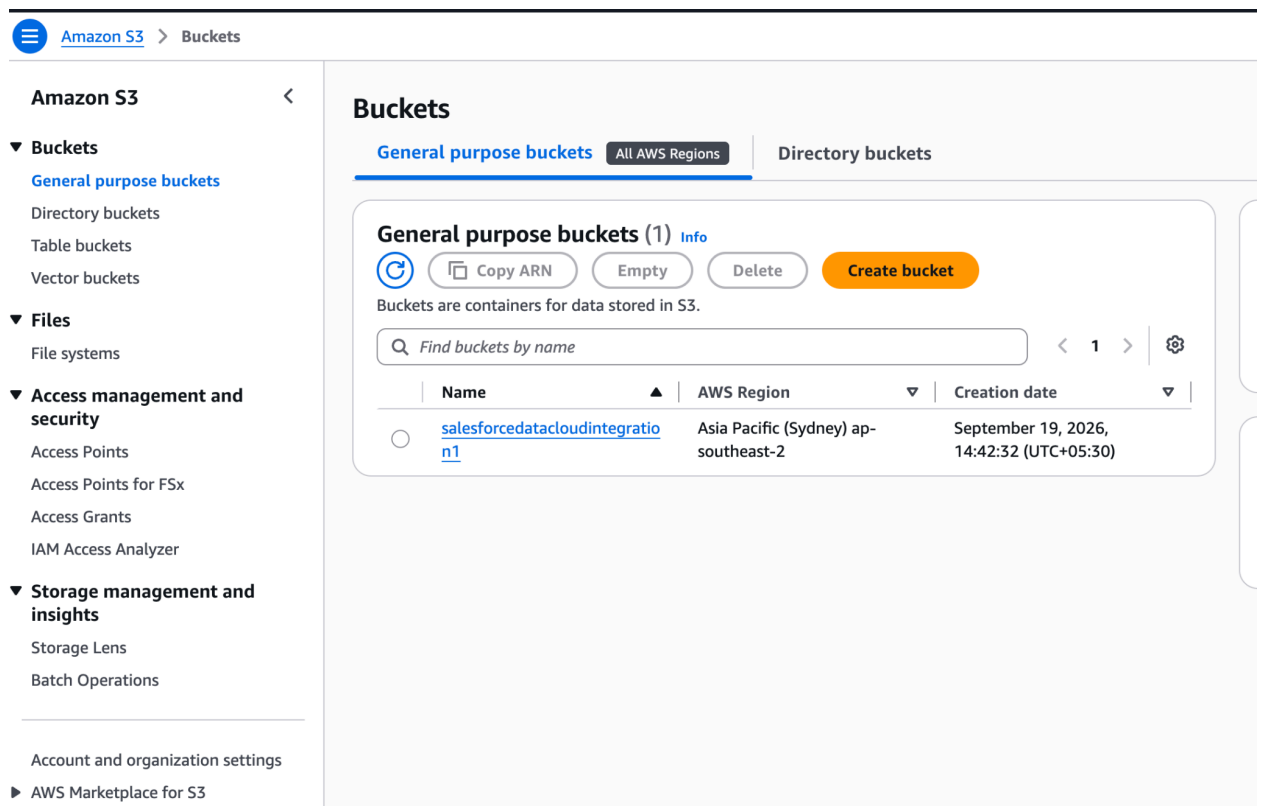
Overview

This document outlines the end-to-end setup for integrating transaction data from **AWS S3** with **Salesforce CRM** contact records using **Salesforce Data Cloud**. The objective is to unify customer profiles, compute custom total order counts, and aggregate order amounts per sales user via Identity Resolution.

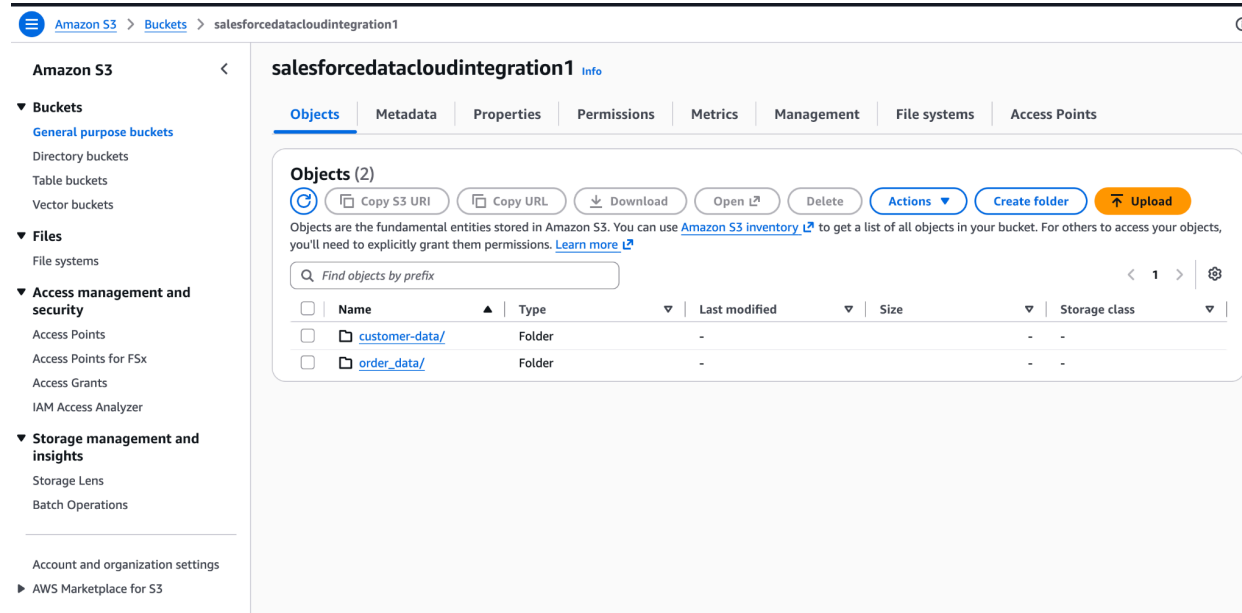
1. AWS Infrastructure Setup

1.1 Storage Provisioning (Amazon S3)

1. **Bucket Creation:** Provisioned Amazon S3 bucket named `salesforcedatacloudintegration1`.

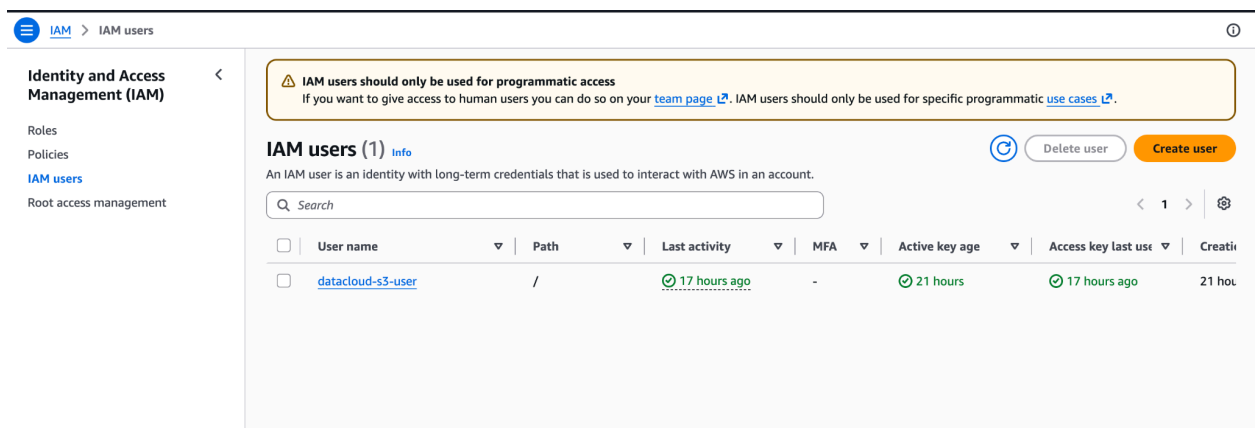


2. **Data Structure:** Structured the bucket to host two primary dataset files in CSV format:
 - **Customer Dataset:** `Customer.csv` (contains core customer profile data).
 - **Order Dataset:** `Order.csv` (contains line-item transactional history).

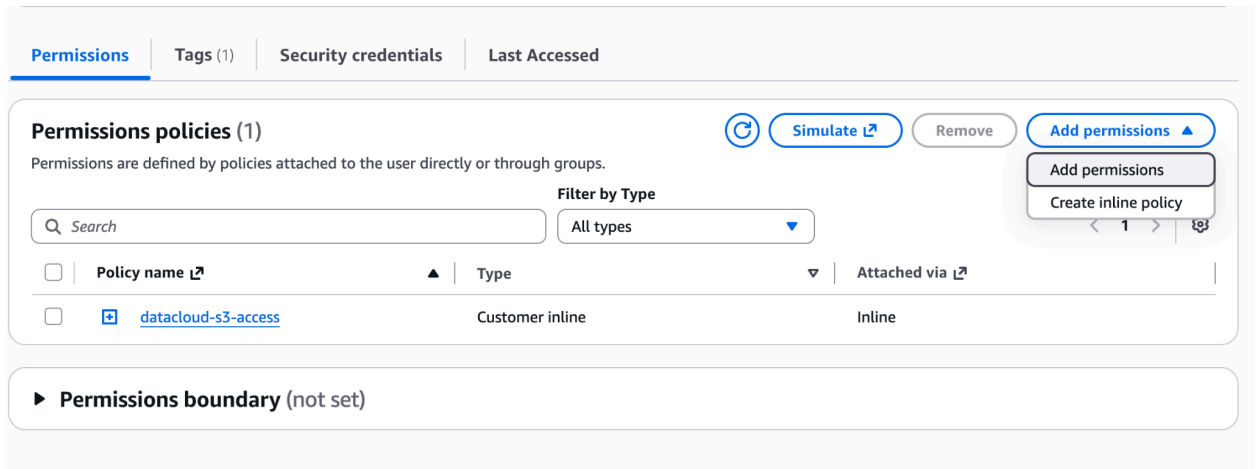


1.2 Access Management (AWS IAM)

1. Created a dedicated IAM user named `datacloud-s3-user` to facilitate Data Cloud authentication.



2. Generated **Access Key ID** and **Secret Access Key** under the user's *Security Credentials* tab.

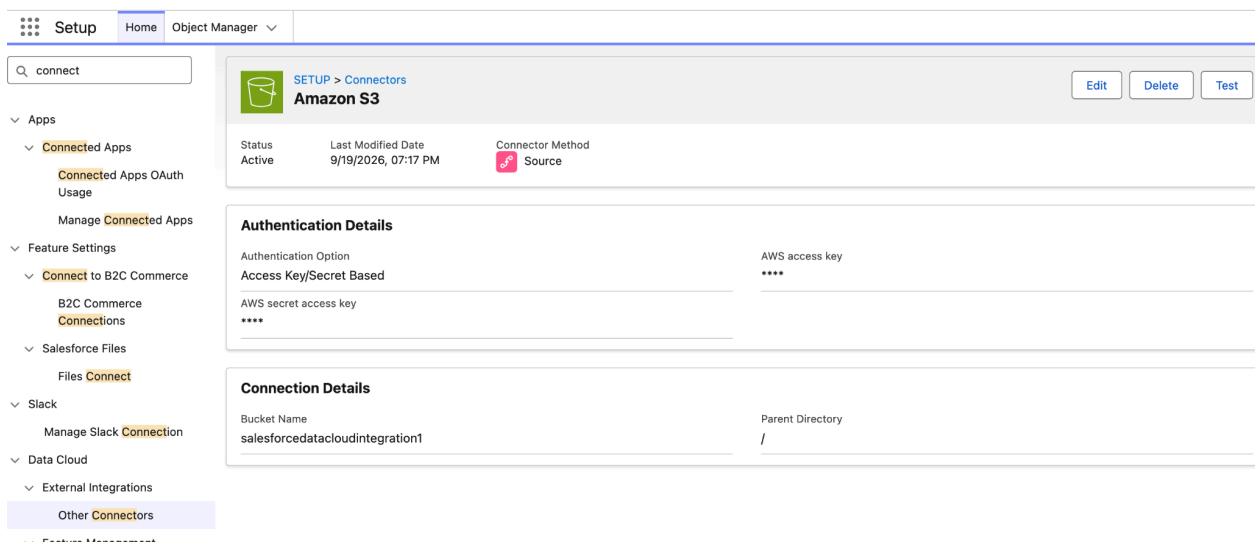


3. Configured an IAM policy granting read-access permissions scoped specifically to `salesforcedatacloudintegration1`.

2. Salesforce Data Cloud Configuration

2.1 Ingestion & Connectivity

- **AWS S3 Connector:** Established the Amazon S3 connector in Data Cloud *Setup* using the IAM user's credentials (`Access Key`, `Secret Key`) and target bucket name.



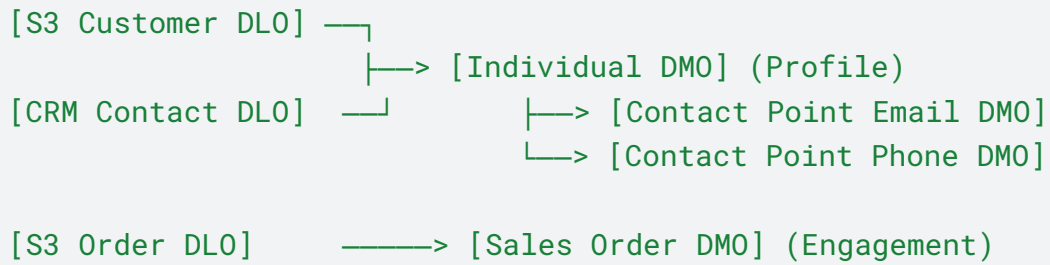
- **Salesforce CRM Connector:** Configured the native CRM connector to ingest standard `Contact` records into Data Cloud.
- **Data Lake Objects (DLOs):** Ingestion generated three core DLOs:
 - `customer_datas3_customer_datacsv_Amazon_dll` (from S3)
 - `order_datas3_order_datacsv_Amazon_S3_dll` (from S3)

- `Contact_Home_dll` (from Salesforce CRM)

3. Data Mapping & Schema Design

Data Lake Objects were mapped to standard Data Cloud **Data Model Objects (DMOs)** based on domain categorization:

None



Data Lake Object

customer-datas3_customer_data.csv Amazon

+ Follow

Preview

Manage Tags

Delete

Storage

Local

Category

Profile

Data Lake Object Status

Active

Last Updated On

9/19/2026, 6:58 AM

Total Records

3

Fields

Details

Refresh History

Data Lake Object Definition

Name

customer_datas3_customer_datacsv_Amazon__dll

Label

customer-datas3_customer_data.csv Amazon

Creation Type

Custom

Category

Profile

Status

Active

Linked Data Stream

customer-datas3_customer_data.csv Amazon

Problem Records Data Lake Object Name

PR_customer_datas3_customer_datacsv_Amaz__dll

Tags

Classifications

Data Mapping

Data mappings have ethics, privacy, and consent considerations. [Learn more about ethics, privacy, and consent in Salesforce Help](#)

Only mapped fields or objects with relationships can be used for segmentation and insights

Data Space

default

Fields mapped

9/10

READY

Review

Data Tagging

Tags help categorizing sensitive data and creating policies to protect

Data Lake Object

order_datas3_order_data.csv Amazon S3

+ Follow

Preview

Manage Tags

Delete

Storage

Local

Category

Engagement

Data Lake Object Status

Active

Last Updated On

9/19/2026, 7:01 AM

Total Records

4

Fields

Details

Refresh History

Data Lake Object Definition

Name

order_datas3_order_datacsv_Amazon_S3__dll

Label

order_datas3_order_data.csv Amazon S3

Creation Type

Custom

Category

Engagement

Status

Active

Linked Data Stream

order_datas3_order_data.csv Amazon S3

Problem Records Data Lake Object Name

PR_order_datas3_order_datacsv_Amazon_S3__dll

Tags

Classifications

Fields (13)

Field Label

Field API Na...

Data Type

Field Used As

Key Qualifier

Tags

Classificatio...

1	cdp_sys_Partitio...	cdp_sys_PartitionD...	DateTime			
2	cdp_sys_Source...	cdp_sys_SourceVer...	Text			

Data Mapping

Data mappings have ethics, privacy, and consent considerations. [Learn more about ethics, privacy, and consent in Salesforce Help](#)

Only mapped fields or objects with relationships can be used for segmentation and insights

Data Space

default

Fields mapped

9/13

READY

Review

Data Tagging

Tags help categorizing sensitive data and creating policies to protect your data. [Learn More](#)

AI is utilized to suggest tags for objects and fields. Only approved tags will be attached to them.

Profile Data Mapping

- **Individual DMO:** Mapped key demographic and identity fields from both **Customer_Info_DL0** and **Contact_CRM_DL0**.
- **Contact Point DMOs:** Mapped phone numbers and email addresses to **Contact Point Phone** and **Contact Point Email** DMOs respectively to handle contact attributes properly.

Engagement Data Mapping

- **Sales Order DMO:** Mapped **Order_Info_DL0** as **Engagement** data to record transactional activities tied to customer profiles.

4. Identity Resolution & Data Reconciliation

To unify customer profiles across AWS S3 and Salesforce CRM:

1. **Match Rules:** Configured custom match rules using **Email Address** as the primary unique identifier to link disparate S3 customer records and CRM contact records.
2. **Reconciliation Rules:** Applied conflict-resolution logic to determine field precedence whenever duplicate or overlapping records match across systems.
3. **Unified Data Output:**
 - **Unified Individual DMO (UniqueIdentifier):** Stores consolidated, deduplicated customer profile records.
 - **Unified Link Object (UniqueIdentifierLink):** Maintains traceability by mapping the primary **UnifiedId** to original source system IDs (**Source Record Keys**).

search...

Data Cloud

Home

Data Streams

Data Lake Objects

Data Model

Data Governance

Data Explorer

AI Models

Semantic Layer

Identity Resolution

Profile Explorer

More

Identity Resolution

Customer_Unique

+ Follow

Update Stats

Edit

Run Ruleset

Data Space

default

Primary Data Model Object

Individual

Ruleset ID

2

Ruleset Status

Published

Last Job Status

Succeeded

Last Job Completed

9/19/2026, 8:42 AM

Ruleset Properties

Details

Processing History

Filters

Match Rules

Custom Rule

Edit

Reconciliation Rules

Matched data sources may provide different values for a single field. Reconciliation rules specify how to select the best value to save to the unified profile. Review default rules and modify as needed. To update multiple fields, use Update Selected.

Contact Point Address

Contact Point Email

Contact Point Phone

Individual

Resolution Summary

Warnings (5)

Total Unified Profiles

965

/ 967 Source Profiles

Consolidation Rate

0%

The amount by which source profiles were combined to produce unified profiles

Known Unified Profiles

965

The number of unified profiles made up of at least one known source profile

Anonymous Unified Profiles

0

The number of unified profiles made up of only anonymous source profiles

Post

Poll

Question

Configure Match Criteria

Configure at least one match criterion. Values in the specified fields will be compared for matches.

*Match Rule Name

Custom Rule

Match Criteria

Data Model Object	Field	Match Method	Advanced Settings
Contact Point Email	Email Address	Exact	Configure

+ Add Criteria

Cancel

Next

You can query on Sales order DMO record to get Information about total order amount purchase by customer , which can help agent to target the customer based on their purchase history

Include Related Objects

Run Query

Help Resources

```
1 SELECT
2     IdentityLink.UnifiedRecordId__c AS UnifiedIndividualId__c,
3     COUNT(SalesOrder.ssot__Id__c) AS Total_Orders_Placed__c,
4     SUM(SalesOrder.ssot__TotalAmount__c) AS Lifetime_Order_Value__c
5 FROM
6     ssot__SalesOrder__d1m SalesOrder|
7 JOIN
8     UnifiedLinkssotIndividual2__d1m IdentityLink
9     ON SalesOrder.ssot__SoldToCustomerId__c = IdentityLink.SourceRecordId__c
10 GROUP BY
11     IdentityLink.UnifiedRecordId__c
```

Query Result

Duration: 46 ms Rows Processed: 128 Rows Retrieved: 3

	UnifiedIndividualId__c	Total_Orders_Placed__c	Lifetime_Order_Value__c
1	8602c4417dfc540eac32ebc9a1a3f349	1	15000
2	ac0aca6334fe49d46218eaf8e0a592e9	1	8900
3	554618dd748a268b2697e42bccaef612	2	16700.5